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A leveraged trading platform for prediction market index baskets.

XLEUTH

A prediction market index

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Disclaimer

Abstract

Prediction markets turn opinions about future events into tradable prices, yet every market remains an isolated bet on a single outcome. There is no widely used instrument that tells a trader, in one number, what the prediction market as a whole believes about a theme or a time window, and no way to trade that number.

Xleuth is a leveraged trading platform for prediction market index baskets. Each index tracks the volume-weighted average price (VWAP) of a basket of Polymarket markets that resolve in the same time bucket, scaled to a 0–100 range. The index is traded as a cash-settled, dated futures contract with up to 5x leverage, and at expiry it settles to the volume-weighted fraction of component markets that resolved YES.

This paper tells the story behind the project, introduces the financial concepts it builds on (prediction markets, options and digital options), specifies the index methodology and contract mathematics, and describes the system that implements them: a data-ingestion pipeline, an EIP-712 signed oracle, an off-chain matching engine, and four smart contracts on Polygon. Xleuth is currently live on the Polygon Amoy testnet.

1 Opening

A little back story about me and why prediction markets in themselves. The journey started in October 2024, a couple of days into the US general election. I started paying lots of attention to the election, especially the presidential election. The night the election result was announced, there I was in my dorm room with some of my friends who were also into some crypto/Forex trading… we observed BTC skyrocket to some resistance price of $74,000-ish, just to make all time high a couple of weeks later. It was a memorable night, watching Bitcoin price action and the effect of Trump winning the reelection. I remember holding Pound-Dollar (GBP/USD) short on an account I was trading at the time, making crazy profit due to the volatility and the opportunity the market brought at the time (I blew the account subsequently, don't bother to ask me about it). Some of my other friends were regretting not going all out into $TRUMP coin. I had a share of regrets as well. Remember, the Pound-Dollar I was holding earlier, I closed out the position two days after, price sold into the 10th of January 2025.

I moved on with my life, with the exuberance that Trump's re-election brought. The crypto community started becoming active again: some were shilling coins and projects, especially those targeting US politicians and some others with weird and ugly names. Some were posting profits they made from the trading they did at the time; some were larping and shilling lifestyle, and the cycle goes on and on. Then I came across an article about a French trader that made $84 million off Polymarket on Trump winning reelection. That got me intrigued about Polymarket. Furthermore, I saw a Joe Rogan podcast where he claimed Elon Musk knew the election winner simply by the odds shown on Polymarket which was by far accurate and leading the rest of the polls and news at the time, then I took a dive into the rabbit hole

That started my exploration into the prediction market, and I was fascinated, while my mind was completely engulfed by the idea of the prediction market and profit-making opportunities possible therein. The concept behind prediction markets in itself is pretty straightforward, but how do you make money from it? That is what I was most concerned about. I read about all the strategies that I could at the time and started devising my own form of strategy.

At the end I decided on arbitrage, though a slightly different approach, and that was the beginning of my journey as a prediction market trader. While the approach worked a little, the strategy itself was a disaster and that led to working on other trading strategies. I took a quantitative approach to prediction markets and treated them as a form of time-dated option, because in the real sense it's a form of exotic option (technical definition later).

The more I research the prediction market, the more gaps in the sector are being exposed. This means that the level of flexibility with a plethora of instruments available in the traditional financial market won't be applicable in this type of nascent market just because most of those are not (yet) available. While there is lots of ongoing innovation in the sector, such as a futures derivative layer on prediction markets, a margin layer, leveraged prediction markets and others, I had an idea of trading a prediction market index, same way I trade indices in traditional finance. This also means that I get to enjoy the benefit of trading the average of all the constituent markets rather than a single market. I was even thinking of other complex trading strategies that can be built on top of an index-like prediction market.

Then came the search, looking for a prediction market index platform. No one was building it. That didn't sit well with me, knowing what that would mean for traders or anyone interested or using prediction markets. I made it my purpose to build a prediction market index platform. I simply describe it as the S&P 500 of prediction markets. Everyone (shrewd and veteran traders, portfolio managers, analysts, even news pundits, basically everyone) knows how great an index such as the SPX (S&P 500) is as an index, tool, market gauge and trading instrument. That is the short (or rather long) backstory behind Xleuth.

2 Introduction

2.1 What Xleuth is

The name Xleuth is a twist on *sleuth* (and pronounced as 'sleuth'), someone who tracks things down, which is exactly what the platform does: it tracks prediction markets and distils them into a single signal.

Xleuth is a leveraged trading platform for prediction market indices. To see what that means, take the definition apart piece by piece:

- **Prediction market.** A market where the price of a share expresses the crowd's probability that an event will happen (defined in Section 3).
- **Index.** A basket of many prediction markets combined into one. Xleuth's most fundamental purpose is to track this basket.
- **Trading.** The index can be traded on an open order book, like any other market, rather than merely observed.
- **Leverage.** The platform lets shrewd traders size up and capitalise even more on the opportunities the index presents.

The approach is simple: group as many markets as possible that share the same resolution time bucket, and let traders trade the average of their prices. The result of the index is therefore the result of the average of its component markets. Instead of trading one prediction market and being exposed to one outcome, you trade the average of many, and the payoff depends on the many instead of the single. You do not have to predict every individual market correctly. You only have to be right about the majority of the weighted outcomes, and that has a lot of use cases from both the trader's and the market operator's standpoint.

2.2 The problem

Prediction markets have grown quickly, but the way people use them has not kept pace with the instrument itself:

- **Fragmented signal.** Hundreds of live markets each describe one event. A trader or analyst who wants to know how the market feels about crypto, elections or the economy within a given month has to read dozens of order books by hand.
- **Single-outcome risk.** A position in one market is a bet on one binary event. It is all-or-nothing, with no diversification, and it is exposed to sudden jumps when breaking news arrives.
- **No gauge.** Equity markets have the S&P 500, volatility markets have the VIX, and crypto has dominance and fear-and-greed indicators. Prediction markets have no widely used aggregate benchmark.
- **Limited tools for risk transfer.** Hedging a prediction market position, or hedging a crypto or traditional portfolio against a cluster of events, is awkward when the only instruments are individual binary shares.

- **A high barrier for newcomers.** Choosing which of hundreds of markets to trade, and sizing each one, is intimidating for anyone just getting started.

2.3 The Xleuth solution and its benefits

Xleuth addresses these gaps by turning a basket of prediction markets into one tradable, benchmarked instrument. The benefits follow directly from the design:

Benefit	What it means in practice
A single market gauge	One 0–100 number summarises the market's collective expectation for a theme and time window, updated every 30 seconds.
Diversified exposure	The payoff depends on many outcomes, so no single event decides the result and idiosyncratic noise is averaged down (Section 5.6).
Probabilistic trading	Traders profit from understanding how the basket is most likely to resolve, not from calling every event right.
Leverage and risk tools	Up to 5x leverage, limit, market and stop orders, and stop-loss and take-profit brackets for risk management.
Hedging	A liquid index position can offset exposure to a Polymarket position, or a crypto or traditional-finance portfolio.
Easier onboarding	Newcomers can trade one familiar, index-like product instead of choosing among hundreds of markets.

3 Background: Prediction Markets, Options and the Derivative Layer

Before describing Xleuth itself, this section sets out the financial building blocks it relies on. Readers who already know prediction markets and options can skip to Section 4.

3.1 Prediction markets

A **prediction market** is an exchange on which participants trade contracts whose payoff depends on the outcome of a future event. In the binary form used by Polymarket, each market poses a yes/no question, such as "Will Bitcoin close above a given price in September?". A YES share pays $1 if the event happens and $0 if it does not; a NO share pays the opposite. Because a YES share is worth $1 only if the event occurs, its price between $0 and $1 can be read as the crowd's estimate of the probability: a YES price of $0.62 implies roughly a 62% chance.

Participants who believe the true probability is higher than the price buy YES, and those who believe it is lower buy NO or sell. Trading takes place on a central limit order book (CLOB), so the price is discovered by supply and demand, and it moves as new information arrives. When the event is settled, the market resolves to YES (1) or NO (0) and the shares are redeemed accordingly. This is why the concept is straightforward, while profiting from it is the harder part.

Growth potential. Prediction markets are expanding along several dimensions at once: more event categories (politics, economics, crypto, technology, culture and geopolitics), more participants and liquidity, and more users who treat the prices as a source of information rather than as a game. Media and analysts increasingly quote market-implied odds, and the ecosystem is now building the same layers that grew around traditional markets: futures, margin, leverage and indices. Each new layer makes the underlying markets more useful and more liquid, which in turn attracts more of the next layer. As of mid-2026, global prediction-market trading volume has surged to roughly $24 billion per month across major platforms Kalshi and Polymarket, up from under $5 billion in late 2025. Analysts at Macquarie and Pew Research Center project total annualized volume exceeding $180 billion by year-end 2026, reflecting explosive growth in liquidity and institutional participation.

3.2 Options and option contracts

An **option** is a contract that gives its holder the right, but not the obligation, to buy or sell an underlying asset at a fixed price (the *strike*, K) on or before a future date (the *expiry*). The buyer pays a premium for that right; the seller takes the premium and the obligation. A **call** option is a right to buy, and a **put** option is a right to sell.

Vanilla options are the standard, plain-structure contracts. For an underlying value S_T at expiry, their payoffs are:

$$C_T = \max(S_T - K,\ 0) \tag{1}$$

$$P_T = \max(K - S_T,\ 0) \tag{2}$$

A vanilla option's payoff grows in proportion to how far the underlying finishes beyond the strike (Figure 1a). **Exotic options** are everything that departs from that plain structure, whether through a different payoff, a path-dependent condition or a special trigger. The simplest and most relevant exotic is the digital option.

3.3 Prediction shares as digital options

A **digital** (or binary) **cash-or-nothing option** pays a fixed amount if the underlying finishes beyond the strike and nothing otherwise. With a payout of 1:

$$Payoff = [S_T > K] \tag{3}$$

where the square brackets mean 1 if the statement inside is true and 0 otherwise. A prediction market share has exactly this payoff: it pays $1 if the event occurs and $0 if it does not, and it expires when the market resolves. This is why I describe a prediction market as a form of time-dated exotic option. The price of such an option is the (discounted) probability of finishing beyond the strike; in Black–Scholes terms, the cash-or-nothing call is worth $e^{(-r\tau)} \cdot N(d_2)$, where τ is the time to expiry. Prediction market prices are quoted directly as that probability.

A log-odds view. A convenient way to reason about how a share's price evolves is to model the market's latent state in log-odds space and treat it as an arithmetic (Bachelier-type) random walk with volatility σ. Let $L(t)$ be the distance, in log-odds units, between the latent state and the 50% strike, and let $\tau = T - t$ be the time remaining. The event resolves YES if the state finishes above the strike, so the share price is:

$$P(t) = \Phi\left(\frac{L(t)}{\sigma\sqrt{\tau}}\right), \tau = T - t \tag{4}$$

where Φ is the standard normal cumulative distribution function. Three consequences matter for traders:

- A market sitting exactly at the strike ($L = 0$) stays at 50% regardless of how much time remains.
- A market that is already on the YES side ($L > 0$) drifts upward towards 1 as time runs out; one on the NO side ($L < 0$) decays towards 0. This is the **time decay** that Section 7 describes harvesting.
- News arrives as jumps in L, which move the price discontinuously; the closer to expiry, the larger the jump in price for the same news (Figure 1b).

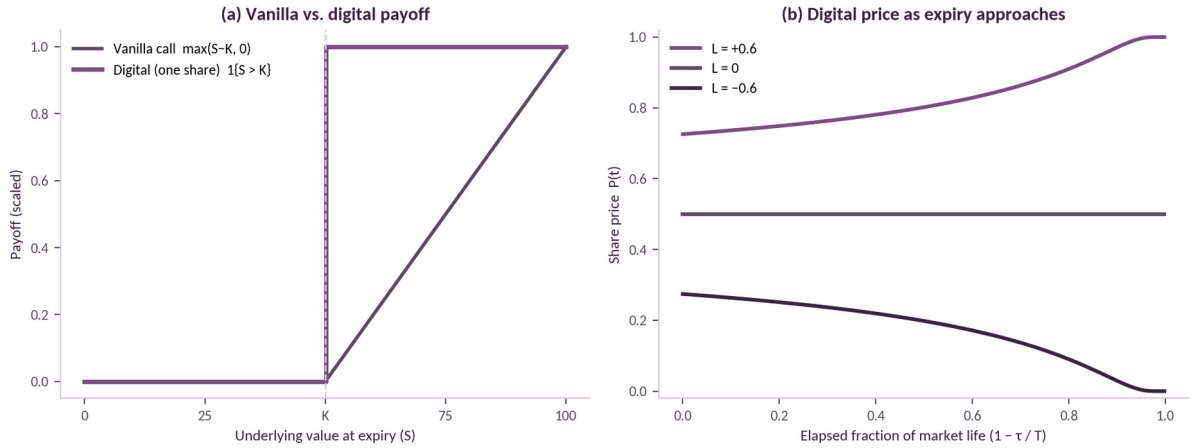


Figure 1. (a) Payoff of a vanilla call versus a digital option. (b) Price of a digital option under the log-odds model for three latent states, with σ = 1. Illustrative.

This framework is used here as an interpretive lens. Xleuth does not price the index with it: the index is observed directly from live Polymarket prices (Section 5).

3.4 The crypto and derivative layer

Polymarket runs on crypto rails: collateral is a dollar stablecoin (USDC) and settlement happens on Polygon. That has two consequences. First, positions are programmable, custody is transparent and settlement is fast and cheap, which is the natural foundation for a derivative layer. Second, the crypto-native trader base is already comfortable with margin and leverage, so the innovations that grew around crypto spot markets are arriving in prediction markets too: futures on prediction market prices, margin layers and leveraged prediction markets.

Nearly all of these innovations still operate on **single markets**. The unexplored space is the layer *above* the single market: an index of many markets, with a derivative on top of it. That is the layer Xleuth builds.

3.5 Prediction market indices

An index condenses many instruments into one number that is easier to read, benchmark and trade than any of its parts. The S&P 500 is the standard example: shrewd and veteran traders, portfolio managers and analysts use it as a market gauge, a benchmark and a trading instrument all at once. A prediction market index does the same for event probabilities. It can answer questions such as “how likely is the crowd to see the markets resolving this month go YES, on average?”, and it can be traded without a view on each individual market.

Three design choices define a prediction market index: which markets are included (the *selection rule*), how they are combined (the *weighting*), and when the index ends (the *resolution window*). Sections 4 and 5 specify the Xleuth answers: selection by theme and volume with sports excluded, volume-weighted averaging, and a shared resolution time bucket.

4 Platform Description

Xleuth tracks the volume-weighted average price of its component markets. Each index has a **ticker symbol**, and the component markets that make up the index are attached to it.

Xleuth allows speculative trading on the indices it tracks. The index is traded as a **cash-settled futures contract with a specified expiration date**, with the distinction that the expiration date is set specifically from the common resolution time of the component markets rather than from a standard exchange calendar.

4.1 Index naming convention

An index name has three parts. The first describes the theme or selection method of the basket, the middle gives the number of markets within the index, and the last gives the expiration or resolution period. For example, **TOP20-NOVEMBER** is the top 20 markets by volume that resolve in November. In the database, each index also has a machine-friendly slug of the form `top20-2026-11`. A compact, futures-style month code (for example `TOP20-U26` for September 2026) is under consideration as a shorter resolution-time notation.

4.2 Why Polymarket

The component market data comes from Polymarket, so Xleuth is a Polymarket index platform. The reasoning is straightforward: Polymarket is the largest prediction market in the world, the Nasdaq of prediction markets. Being the largest means liquidity backs each market, which helps individual market price discovery. Furthermore, because of that depth, there is little difference in price between Polymarket and other prediction markets, so Polymarket is the platform of choice for a reference price.

4.3 Data sources

For every cycle, Xleuth combines three public Polymarket interfaces, each used for what it does best:

Interface	Used for
Gamma API	Market discovery: question, category, end date and the identifiers of the outcome tokens.
CLOB API	Live prices: the midpoint of the YES token, requested in batches.
Data API (live volume)	Live trading volume per market, scoped by event.

Cached or reference prices from the discovery interface are never used to compute the index. They serve only as a fallback when a live midpoint is unavailable.

5 Methodology and Index Mathematics

As defined earlier, Xleuth tracks the volume-weighted average price (VWAP) of the component markets. This section specifies how markets are selected, how they are grouped in time, and exactly how the index is computed and settled.

5.1 Constituent selection

Markets are selected by the theme of the index. They cut across the sectors and sub-sectors on Polymarket, specifically finance, crypto, tech, politics and geopolitics, culture and entertainment, elections and global elections, economy and weather, but not limited to these. For each index the selection procedure is:

1. Take all active, unclosed markets, ranked by volume.
2. Keep only markets whose end date falls inside the index's resolution window (the temporal bucket).
3. Exclude the excluded categories (sports by default).
4. Keep the top N markets by volume, where N is the basket size.

Why sports markets are excluded

Sports markets are generally excluded. While they carry a lot of volume, that volume introduces two risks for index construction:

- **Weight concentration.** Because the index is a volume-weighted arithmetic mean, huge volume from the sports sector can skew the calculation. The concluded FIFA World Cup is an example: the tournament-winner market alone reached a volume of $4.3 billion, and many other sports markets show the same effect. A single market of that size would dominate the weights and turn the index back into a single bet (Section 5.6).
- **Jump discontinuity.** Sports prices jump when breaking news arrives, such as a goal, an injury or a red card. Such sudden inflows of information can skew the index wrongly.

The index tracks prediction markets and gauges market sentiment for the theme, but it can also be used as a speculative vehicle, and we do not want it to be seen as a pure gambling vehicle. That is the main reason the sports sector is entirely excluded. One can argue that pop culture and entertainment offer the same risk as sports, but they offer lesser risk, and there is a legal and regulatory reason for the exclusion as well.

5.2 Temporal bucketing

Indices are bucketed together by similar resolution time. Take **TOP20-SEPTEMBER**, for example: the top 20 markets by volume that have a resolution time in September 2026. The exact date does not matter; what matters is whether the date falls within the range of the index. If one market resolves on 17 September and another on 25 September, both resolution dates fall within September 2026 and therefore belong to the same index within the thematic basket. The expiry of the tradable contract is set from this window.

5.3 Price and volume inputs

For each constituent market *i*, every cycle (30 seconds) produces two inputs:

- **Price** $p_i \in [0, 1]$: the midpoint of the YES token from the CLOB. If the midpoint is unavailable, the last reference price from the discovery interface is used as a fallback.
- **Volume** $v_i > 0$: the live volume of the market, taken from the event-scoped volume interface.

Several markets can belong to the same Polymarket *event*, and the volume interface reports at the event level. To avoid double counting, Xleuth requests each event once, de-duplicates by market identifier, and assigns each market only its own volume. Markets with no price or no volume are skipped.

5.4 Index calculation

The index is the volume-weighted average price of the N constituent markets. Writing the weight of market *i* as its share of total volume,

$$\omega_i = \frac{v_i}{\sum_{i=1}^{N} v_i} \tag{5}$$

the index level in probability space and on the published 0–100 scale are:

$$I_t = \frac{\sum_{i=1}^{N} v_i p_i}{\sum_{i=1}^{N} v_i} = \sum_{i=1}^{N} \omega_i p_i, S_t = 100 \cdot I_t \tag{6}$$

This simply means that the larger the share of total volume a market has, the more weight its price carries in the index price. The implementation is the one-line function `vwap.basket_vwap([(price, volume)]) = sum(p·v) / sum(v)`.

Markets with sub-outcomes. Some markets have no single Yes/No question. “What price will BTC reach in September 2026?” is an example: a range ($67,500–$70,000, $70,000–$72,500, and so on) or a single price ($70,000, $75,000, and so on) is offered, and each sub-component carries its own binary Yes/No market. Such a market *m* with sub-outcomes *j* is first VWAPed internally into a single price and volume, and then enters the global calculation like any other market:

$$p_m = \frac{\sum_j w_{m,j} q_{m,j}}{\sum_j w_{m,j}}, v_m = \sum_j w_{m,j} \tag{7}$$

where q is the price and w the volume of sub-outcome j. Once markets with sub-components are VWAPed internally, the global VWAP provides the single price point that is tracked.

A worked example

Take four constituent markets with the illustrative prices and volumes below (volume in $ millions):

Market	Price p	Volume v	Weight ω	v × p
A	0.80	40	0.40	32.0
B	0.30	30	0.30	9.0
C	0.55	20	0.20	11.0

Market	Price p	Volume v	Weight ω	v × p
D	0.10	10	0.10	1.0
Total		**100**	**1.00**	**53.0**

The index is I = 53.0 / 100 = 0.53, so the published level is **S = 53.0**. A simple unweighted average of the four prices would give 43.75, which shows how much the volume weighting changes the result. Now suppose A and C resolve YES and B and D resolve NO. The settlement value is R = (40 + 20) / 100 = 0.60, so the contract settles at 60.

5.5 Resolution and settlement value

Polymarket resolves each market to YES (price 1) or NO (price 0). When a market resolves, that final state is reflected in the index: a YES market contributes price 1 and a NO market price 0, and it no longer carries information about the future. Once some constituents start resolving, the index keeps running until the last market has resolved.

At expiry, every open position is settled to the **realised fraction**: the volume-weighted share of constituents that resolved YES, where $y_i = 1$ if market i resolved YES and 0 otherwise.

$$R = \frac{\sum_{i=1}^{N} v_i y_i}{\sum_{i=1}^{N} v_i}, S_T = 100 \cdot R \tag{8}$$

If every constituent resolves YES the index settles at 100; if every constituent resolves NO it settles at 0; and if half of the weighted volume resolves YES, it settles at 50. Settlement uses the same de-duplicated volumes as the live index, never raw cached volumes.

5.6 Index properties

The index is an unbiased forecast of its own settlement. If market prices are probabilities, then $E[y_i] = p_i$, and with the weights held fixed the expected settlement value equals the index level:

$$E\left[R|\mathcal{F}_t\right] = \sum_{i=1}^{N} \omega_i p_i = I_t \tag{9}$$

In practice weights drift as volumes change, so this is an approximation rather than an exact martingale property. It does explain why the index is a meaningful gauge: it is the market-implied expected value of the realised fraction.

Diversification and concentration. Assuming the component outcomes are independent, the variance of the settlement fraction is

$$Var(R) = \sum_{i=1}^{N} {\omega_i}^2 p_i \left(1 - p_i\right) \tag{10}$$

Because ω_i^2 is small when weights are spread out, the variance falls as the basket becomes more diversified. A convenient measure of how diversified the basket really is, is the effective number of markets:

$$N_{eff} = \frac{1}{\sum_{i=1}^{N} \omega_i^2} \tag{11}$$

For N equal-weight markets, N_{eff} = N. If one market holds almost all the volume, N_{eff} approaches 1 and the index degenerates into a single bet. This is the mathematical reason sports markets are excluded, and it suggests a useful publishing metric: each index page can display its N_{eff} alongside its level. Correlated components (for example, several markets on the same election) add positive covariance terms and reduce the diversification benefit further.

Bounded range. Because the index lives on a 0–100 scale, the maximum loss on a position is bounded: a long entered at price E can lose at most E per unit, and a short at most 100 − E. This bound is what makes high leverage tractable at the contract level (Section 6).

5.7 Index characteristics

The index opening price is not important in itself. It depends on when the index is created and how close that date is to the creation of the underlying Polymarket markets, and, above all, on the starting prices at index creation, which in turn depend on how the markets are seeded, the liquidity available and the volume being traded. The opening price can start anywhere on the scale.

The beauty and elegance of this design is that you do not have to get every prediction right. You only have to understand, probabilistically, how the index is most likely to resolve, and you can capitalise on it.

6 The Index Derivative Layer

While Xleuth tracks multiple indices, it also allows speculative capitalisation of the opportunity they present, in an innovative way. The derivative layer functions as a **cash-settled futures contract with a specified expiration date**, where the expiration date follows the temporal bucket described in Section 5.2.

The derivative layer is a **linear derivative of the index price**, which lets traders set bid and ask prices and make a market according to participant supply and demand. The underlying markets are a form of exotic cash-or-nothing option, but the traded layer is linear in the index, so it needs no separate pricing model: the pricing model is already baked into the Polymarket data that drives the index.

6.1 Contract specification

Parameter	Value
Underlying	Volume-weighted average of the YES prices of the basket's constituent Polymarket markets
Quote scale	0 to 100 index points
Tick size	0.01
Contract type	Cash-settled, dated future; expiry follows the basket's resolution window
Settlement value	Realised volume-weighted YES fraction, scaled to 0–100 (Section 5.5)
Collateral	USDC (test USDC on Amoy)
Maximum leverage	5x
Minimum order size	10
Maintenance margin	6.25% of notional (625 basis points)
Liquidation fee	0.5% of notional (50 basis points), paid to the liquidator
Trading fee	10 basis points (reporting only in version 1; not yet charged)
Order types	Limit, market and stop; stop-loss and take-profit on open positions

6.2 Order types

The trading method allows three order types. **Limit** and **stop** orders follow a good-till-cancelled (GTC) policy, which means the order stays open until it is cancelled. **Market** orders follow a fill-or-kill (FOK) policy with a maximum-slippage guard: the order either fills completely within the trader's accepted worst price or is rejected. A **stop** order is a standalone conditional order that converts to a market order when the mark price crosses its trigger, which is useful for breakout entries. Pending limit and stop orders are reflected in the order book, which is integrated into the design; the order book can be previewed and the available liquidity analysed before making a trading decision.

For an open position, a trader can attach **stop-loss and take-profit** orders for risk management. They work as a bracket: one-cancels-the-other, closing the full position, and triggered by the index mark price.

6.3 Margin and position accounting

Let a position have size q, side s (+1 for long, −1 for short), entry price E and leverage $\lambda \in [1, 5]$. The margin locked when the position is opened or increased at oracle price P is

$$M = \frac{P \cdot q}{\lambda} \tag{12}$$

Margin is always computed from the oracle price and not from the limit price, so a trader cannot reduce collateral requirements by posting a distant limit price. When a position is increased on the same side, the entry price becomes the size-weighted average:

$$E' = \frac{E \cdot q + P \cdot \Delta q}{q + \Delta q} \tag{13}$$

The unrealised profit and loss, and the position's equity, at the current mark price P are

$$U = s \cdot (P - E) \cdot q, Equity = \max(M + U,\ 0) \tag{14}$$

At settlement, the same formula applies with the final settlement value S_T in place of P. A long position therefore gains when the index finishes above its entry price, and a short position gains when it finishes below. Because losses are floored at the locked margin, a position's payout can never be negative.

6.4 Liquidation

A position is liquidated when its equity falls below the maintenance margin. Writing m for the maintenance ratio (6.25%) and f for the liquidation fee rate (0.5%), a position is liquidatable if and only if

$$Equity < m \cdot P \cdot q \tag{15}$$

The comparison is strict: equity exactly at the threshold is not liquidatable. When a position is liquidated, the liquidator receives a reward equal to the smaller of the fee and the remaining equity, and the rest is returned to the trader:

$$Reward = \min(f \cdot P \cdot q, Equity) \tag{16}$$

Solving the liquidation condition for price, with margin locked at entry ($M = E \cdot q/\lambda$), gives the liquidation price for each side:

$$P_{liq,long} = \frac{E\left(1 - \frac{1}{\lambda}\right)}{1 - m} \tag{17}$$

$$P_{liq,short} = \frac{E\left(1 + \frac{1}{\lambda}\right)}{1 + m} \tag{18}$$

For example, a long entered at 60 with 5x leverage is liquidated when the oracle price falls below $60 \times 0.8 / 0.9375 \approx 51.2$. A short entered at 40 with 5x is liquidated above $40 \times 1.2 / 1.0625 \approx 45.2$. Because

the index cannot exceed 100, a short entered above roughly 88.5 at 5x has a liquidation price above the top of the scale and cannot be liquidated by price movement alone.

Liquidation is permissionless: anyone can trigger it on-chain once the condition holds, and the oracle price used must be fresh.

6.5 Settlement at expiry

When the last constituent market resolves and the basket has expired, the realised fraction is computed off-chain from the de-duplicated live volumes and signed by the settlement signer. Any relayer can then submit the signed value to the settlement contract, which verifies the expiry, the signature and its freshness, and settles every open position at that value exactly once. Each position's collateral is released as max(M + U, 0), with U computed against the settlement value (Section 8.5).

7 Index Use Cases

As described earlier, the index derivative layer can be used for speculative trading, by allowing the index to be traded as a cash-settled futures contract whose expiration is the resolution date. The index price is scaled from 0 to 100, the tick size is 0.01, and the maximum leverage available is 5x.

7.1 Arbitrageur paradise

As a new project just starting out, Xleuth offers many opportunities for arbitrageurs, owing to gross mispricing by market participants at this early stage and a lack of liquidity. Thin liquidity is one thing; another big factor is how mispriced the index trading price can be as part of the price discovery process. This creates two main kinds of arbitrage.

1. **Market rebalancing arbitrage.** Arbitrageurs capitalise on the drift, or price differential, between the index's trading price and its underlying index price. If the trading price is higher than the underlying index price, they sell the index, forcing it back to the actual price; if it is lower, they buy the index, bringing in demand and forcing the price back up. They act as the **rebalancing force** that keeps the traded price aligned, and this type of arbitrageur is essential to the operation of Xleuth.
2. **Replication arbitrage.** Arbitrageurs replicate the index's component markets on Polymarket while hedging the price differential on Xleuth. This is similar to pair trading, whereby the relationship between the component markets and the index is capitalised on. These are methods of making speculative profit from index trading without strictly predicting the market or relying on prediction market resolution.

Both strategies can be written down precisely. Let F_t be the traded price of the Xleuth contract and S_t the index level. The **basis** is

$$b_t = F_t - S_t \tag{19}$$

At expiry the contract settles to S_T, so the basis converges to zero. In between, the index can be replicated on Polymarket by buying YES shares in each constituent in proportion to its weight. Holding

$$n_i = 100 \cdot q \cdot \omega_i \quad shares\ of\ market\ \ i \tag{20}$$

costs $\Sigma\, n_i \cdot p_i = q \cdot S_t$ and pays $\Sigma\, n_i \cdot y_i = q \cdot S_T$ at resolution, which is exactly the payoff of q index contracts. When the basis is positive (F above S), an arbitrageur sells the contract and buys the replicating basket; when it is negative, the reverse. The profit is locked in by convergence and does not depend on how the markets resolve, subject to trading costs, liquidity, and the fact that volume-based weights drift over time and need rebalancing.

7.2 Other ways to use the index

Beyond the arbitrageur's toolkit, the index supports a range of strategies. This list is not exhaustive:

1. **Hedging prediction market positions.** Hedging a position on Polymarket or any other prediction market.

2. **Hedging crypto or traditional finance portfolios.** Offsetting exposure to a cluster of events, such as a crypto-themed or economy-themed index, against a broader portfolio.
3. **Option-style time-decay harvesting.** Modelling the underlying markets as options and harvesting their time decay as they approach resolution (Section 3.3).
4. **Directional position trading with leverage.** Capitalising on moves in the index with leverage, which can be very attractive to native crypto and prediction market traders. This is purely speculative trading.
5. **Market making.** Providing liquidity to the order book and earning the spread.
6. **Algorithmic trading.** Latency arbitrage between the live Polymarket data and the traded price, and many other systematic strategies.
7. **Sentiment and research.** Using the index history as a benchmark of thematic market sentiment, by analysts, news desks and researchers who do not trade it at all.

8 System Architecture and Implementation

Xleuth is built as a small set of focused services around four smart contracts. Off-chain services handle what must be fast and data-heavy (reading Polymarket, computing the index, matching orders); on-chain contracts handle what must be trustworthy (custody of collateral, positions, price feeds and settlement). All services share one library that defines index naming, lifecycle rules and data schemas, so the rules live in one place.

8.1 Architecture overview

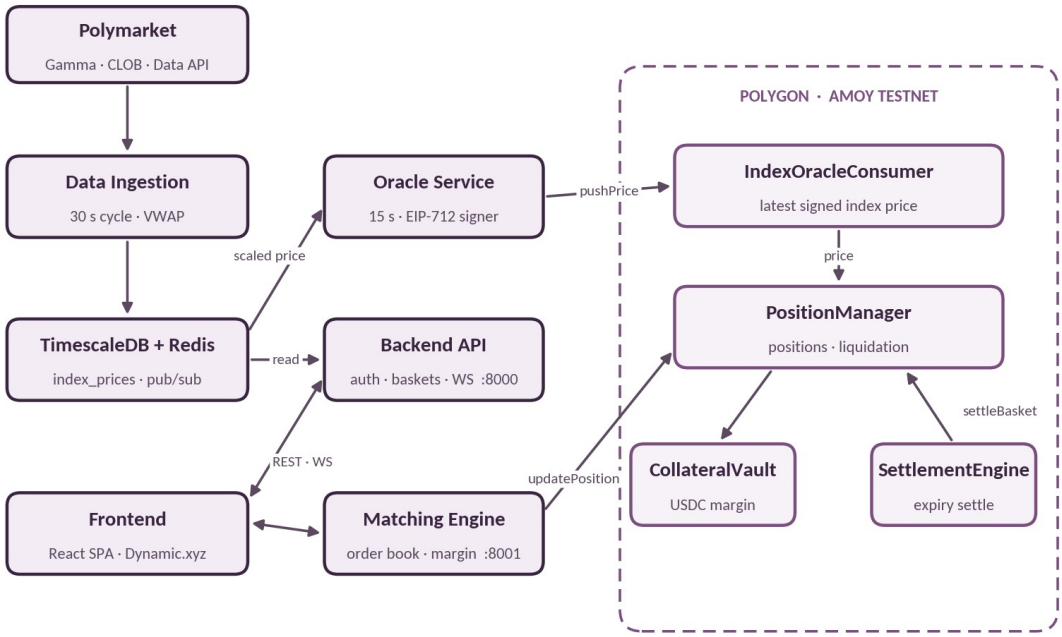


Figure 2. Xleuth system architecture. Off-chain services (left) and on-chain contracts (right).

Component	Responsibility
Data ingestion	Polls Polymarket every 30 seconds, selects constituents, computes each basket's VWAP and stores it.
TimescaleDB and Redis	Time-series storage of index prices and constituent prices; Redis publishes live prices and order books.
Oracle service	Reads the latest index price every 15 seconds, signs it and pushes it on-chain.
Backend API	Authentication, basket data, price history and candles, account data, test-token faucet and live WebSocket feeds.
Matching engine	Order book, margin checks, stop and bracket triggers, and submission of matched fills for on-chain settlement.
Smart contracts	Collateral custody, positions and liquidation, the signed price feed, and expiry settlement.

Component	Responsibility
Frontend	A React single-page application with an interactive price chart, order book, trading ticket and portfolio.

8.2 Data ingestion

Every 30 seconds the ingestion service takes a single snapshot of the active Polymarket markets, ranked by volume, and uses it to update every active index. For each index it selects the constituents (Section 5.1), reads YES midpoints from the CLOB in one batch, fetches de-duplicated live volumes by event, and computes the VWAP. Results are written to time-series tables and published to Redis so the API can stream them to users. Index definitions are data, not configuration: the service reads the active indices from the database, so a newly activated index is picked up on the next cycle without a restart. Expired indices are kept as history and never deleted.

8.3 Oracle

The oracle service reads the latest scaled index price every 15 seconds. It rounds the price to the 1–100 scale used on-chain, signs an EIP-712 typed message and submits it to the price-feed contract:

$$\hat{S} = \min\left(100, \max\left(1, round\left(S_t\right)\right)\right) \tag{21}$$

The signed message has the form `PriceUpdate(basketId, price, timestamp)` under the domain “Xleuth Index Oracle”, where `basketId` is the keccak-256 hash of the index slug. Failed submissions are retried with exponential backoff, and an update identical to the one already on-chain is skipped. The oracle only pushes for active, registered indices, and stops after expiry. In the current testnet build, the on-chain reference price is an integer on the 1–100 scale, while the off-chain index keeps two decimals.

8.4 Matching engine

Orders are matched **off-chain** and settled on-chain. This is a deliberate design decision: an order book with price-time priority is far cheaper and faster off-chain, and avoids the failure modes of pool-based designs, while custody and position accounting stay on-chain where they can be verified. The engine is built on a proven order-matching library and adds the Xleuth-specific logic:

- **Order book.** Price-time priority, aggregated depth, and a guard against self-trading.
- **Market orders.** Fill-or-kill with a maximum-slippage bound: the engine computes the worst acceptable price, checks that enough depth exists inside it, walks the book level by level, and otherwise rejects the order with no partial fill.
- **Stops and brackets.** Stop orders and stop-loss/take-profit brackets are evaluated against the oracle mark price every five seconds; a triggered bracket cancels its sibling.
- **Pre-trade margin check.** Before an order is accepted, the engine validates the basket, size, leverage, tick alignment, oracle freshness (no older than 60 seconds), the on-chain leverage cap, and that free collateral covers the required margin, computed at the oracle price.

- **On-chain settlement of fills.** Fills are netted per trader and basket and submitted to the position contract. An order is only confirmed if **both** the taker's and maker's legs are mined; otherwise both are rejected, so the order book never diverges from on-chain state.
- **Idempotency.** Each order carries an idempotency key, so a retried request returns the original result instead of creating a duplicate.

8.5 Smart contracts

Four contracts, written in Solidity 0.8.24 with OpenZeppelin building blocks, hold the trust-critical logic:

Contract	Role
CollateralVault	Holds USDC. Users deposit and withdraw only their *free* balance; margin is moved between free and locked only by the position contract. Settlement credits max(locked + PnL, 0), so a loss is floored at the locked margin.
PositionManager	Keeps one net position per user per index. Opens, increases, reduces, flips and closes positions; locks margin at the oracle price; refuses to act on stale prices; liquidates under-margined positions permissionlessly; and settles all open positions at expiry.
IndexOracleConsumer	Stores the latest signed index price per index. Accepts an update only if its timestamp is newer than the last, not stale, within range (0 to 100 scale) and signed by the authorised oracle (EIP-712).
SettlementEngine	Settles an index exactly once after expiry, from a signed realised-fraction value, by calling the position contract for every open position.

Settlement flow. After expiry, the realised fraction R (Section 5.5) is computed off-chain from de-duplicated live volumes and signed under the domain "Xleuth Settlement". Any relayer can submit it. The settlement contract checks that the index is registered, has expired and has not been settled; that the value lies within range; that the signature is fresh and authorised; and then settles every open position in one pass: the position manager iterates the open positions of the index, computes each PnL against the settlement value, and credits the vault.

8.6 Index lifecycle and administration

Indices are rows in a database table, not hard-coded names. They are created only through an administrative command-line tool, which exposes no HTTP endpoint, so the public API has no path that can create a market. An index moves through a fixed lifecycle:

State	Meaning
draft	Created with a theme, size, window and expiry; constituents previewed by a dry run.
registering	Expiry registration submitted on-chain; waiting for confirmations.
active	Registered on-chain and confirmed. The index is priced, pushed to the oracle and tradeable.
closed	Past expiry; no longer tradeable, awaiting settlement.

State	Meaning
settled	Settlement completed on-chain.

An index never becomes active before the chain confirms its registration. Every transition is recorded in an audit table, and a reconcile command compares the database against on-chain expiry, price and settlement status. Month-end rollover is a routine: preview the next month's index, create it, register it and confirm that it is active and receiving prices before the first of the month.

8.7 Authentication and access

Users sign in with an email or wallet through Dynamic.xyz, which provides embedded wallets. The backend verifies Dynamic's signed token against its published keys and issues a short-lived internal token (60 minutes) that the API and the matching engine accept. Market data is public; account, order and position endpoints require a token and enforce ownership. To try the platform on testnet, a user can claim 1,000 test USDC from a faucet, limited to one claim per wallet every 24 hours.

8.8 Security and risk controls

Layer	Control
Trading	Margin on oracle price; 60-second price staleness limit; tick and minimum-size validation; leverage cap enforced off-chain and on-chain; self-trade rejection; fill-or-kill with slippage; idempotency keys; ownership checks.
Contracts	Owner-only configuration; only the position contract can move margin; only the settlement contract can settle; reentrancy protection on deposits; monotonic, range and signature checks on prices; once-only settlement; loss floored at zero.
API	Per-IP sliding-window rate limits (for example, 30 sign-ins, 10 faucet claims and 120 orders per minute); request-size limit; strict input validation; HTTPS and security headers in production.
Operations	Secrets kept out of example files and checked by automated scripts; separate keys for deployer, oracle, settlement and faucet operator; basket creation restricted to the admin tool.

8.9 Testing and verification

Each layer has its own automated test suite: unit tests for index construction, VWAP and the realised fraction (including a test proving that markets sharing an event are not double counted); order-book tests for price-time priority, fill-or-kill, triggers and brackets; margin, settlement and API tests with simulated chain components; Foundry tests for every contract, including fuzzing; and live end-to-end tests on Amoy that run the entire path: Polymarket data, VWAP, oracle push, matching, expiry, settlement and payout.

9 Blockchain and Tech Stack

The index trading layer is built on **Polygon**, an Ethereum Layer 2. The decision was made strictly out of necessity: Polygon powers Polymarket, the largest prediction market in the world. It is user friendly, with very low gas fees for transactions. We are also working on gas sponsorship for users, which will be available when the project goes live on mainnet.

At the moment, the project is live on the **Polygon Amoy testnet**, using test USDC with no monetary value.

Layer	Technology
Frontend	React 18 single-page application, TypeScript, Vite, Tailwind CSS, Radix UI, TradingView Lightweight Charts, React Query
Wallets and auth	Dynamic.xyz embedded wallets; JWKS-verified tokens exchanged for internal tokens
Backend services	Python 3.12, FastAPI, asyncpg, web3.py, pydantic; shared library across services
Matching	Order-matching library extended with margin, stop and bracket logic
Data	TimescaleDB (PostgreSQL 16) and Redis 7
Smart contracts	Solidity 0.8.24, Foundry, OpenZeppelin, EIP-712 signatures
Chain	Polygon Amoy testnet (chain ID 80002); USDC collateral
Infrastructure	Docker Compose for services; static frontend hosting

10 Risks, Limitations and Trust Assumptions

Xleuth is early, and it is better to state its limits plainly than to hide them. The most important ones are:

- **Oracle and settlement trust.** In the MVP, one authorised signer provides index prices and another provides the settlement value. Users trust that these keys are secure and honest until the signing is decentralised (Section 11).
- **Dependence on Polymarket.** The index inherits Polymarket's data, liquidity and resolution process. API changes, outages, disputed resolutions or delisted markets can affect the index.
- **Basket composition risk.** Volume-weighted indices can still be dominated by a few large markets, and components are often correlated. The effective number of markets (Section 5.6) should be monitored.
- **Leverage.** Leverage of up to 5x magnifies losses as well as gains, and positions can be liquidated. Prices can gap between oracle updates.
- **Representation limits.** The on-chain price and settlement value are integers on a 1–100 scale in the current build, so outcomes at the very extremes are represented at the boundary values.

- **Shortfall handling.** Losses are floored at the locked margin; any shortfall beyond that is absorbed by the platform in this version.
- **Centralised matching.** The order book runs as a single off-chain instance. Fills are verified on-chain, but order availability depends on the operator.
- **Smart contract risk.** The contracts have been through internal review and automated testing, and are deployed on a testnet only.
- **Regulatory risk.** Derivative products on event outcomes may be restricted in some jurisdictions. Sports markets are excluded partly for this reason, and jurisdictional limits will be defined before any mainnet launch.

11 Roadmap

The following directions follow from the architecture and the limits listed above. They are intentions, not commitments, and they carry no dates.

Area	Direction
Mainnet	Move from Amoy to Polygon mainnet with real USDC, deposits and withdrawals enabled, and gas sponsorship for users.
Trust minimisation	Replace single-key signing with multiple independent signers or a threshold scheme for the oracle and settlement values; automate the settlement relay.
Fees and incentives	Switch on trading fees (already measured, not yet charged) and consider liquidity incentives for early market makers.
More indices	Add thematic baskets (for example, a single geopolitical theme) and further time buckets alongside the cross-sector top-volume baskets.
Analytics	Publish index-quality metrics such as effective number of markets, constituent turnover and basis to the benchmark.
Strategies	Build more complex strategies on top of the index, such as spreads between time buckets or themes.
Governance and compliance	Define restricted jurisdictions and governing terms; commission an independent security audit before real funds are used.

12 Conclusion

Prediction markets have made opinions about the future tradable, but one event at a time. The S&P 500 showed what happens when many instruments are condensed into one benchmark: a gauge, a trading instrument and a common language. Xleuth brings that idea to prediction markets. By tracking the volume-weighted average of Polymarket markets in a shared resolution window, and by letting traders take leveraged positions on that average, it offers exposure to the probabilistic shape of a whole theme, rather than to a single outcome.

The design aims to be simple where it can be: one index formula, one contract type, one settlement rule. The implementation is already live on testnet, from the data pipeline through the matching engine to the contracts that settle at expiry. What comes next is hardening, decentralising and opening it up to more traders, and I am glad to build it in the open.

Appendix A: Notation and Formula Summary

Symbol	Meaning
N	Number of constituent markets in an index
p_i, v_i, ω_i	YES price, live volume and volume weight of market i
y_i	Resolution of market i: 1 if YES, 0 if NO
I_t, S_t	Index level in [0,1] and on the 0–100 scale at time t
R, S_T	Realised YES fraction and settlement value ($100 \cdot R$)
N_{eff}	Effective number of markets, $1 / \Sigma\, \omega_i^2$
F_t, b_t	Traded contract price and basis ($F_t - S_t$)
q, s, E, λ	Position size, side (+1 long, −1 short), entry price and leverage
M, U	Locked margin and unrealised PnL
m, f	Maintenance margin ratio (6.25%) and liquidation fee rate (0.5%)
$L(t)$, σ, τ	Latent log-odds distance from the strike, its volatility and time to expiry
Φ	Standard normal cumulative distribution function

Quantity	Formula
Index level	$I_t = \Sigma\, v_i\, p_i / \Sigma\, v_i$; $S_t = 100 \cdot I_t$
Sub-outcome market	$p_m = \Sigma\, w_j\, q_j / \Sigma\, w_j$; $v_m = \Sigma\, w_j$
Settlement	$R = \Sigma\, v_i\, y_i / \Sigma\, v_i$; $S_T = 100 \cdot R$
Digital share price	$P(t) = \Phi(\, L(t) / (\sigma\sqrt{\tau})\,)$
Margin	$M = P \cdot q / \lambda$
Unrealised PnL	$U = s \cdot (P - E) \cdot q$
Liquidation test	$\max(M + U, 0) < m \cdot P \cdot q$
Liquidation price	Long: $E(1 - 1/\lambda)/(1 - m)$; Short: $E(1 + 1/\lambda)/(1 + m)$
Replicating holdings	$n_i = 100 \cdot q \cdot \omega_i$ shares of market i

Appendix B: Testnet Deployment and Default Parameters

Contracts deployed on Polygon Amoy (chain ID 80002). These are testnet contracts using test funds with no value, and the deployment may be reset.

Contract	Address
CollateralVault	`0x6dbeEb6a5bE0cD63159b1dBD5525dE125A1CECef`

Contract	Address
PositionManager	`0x65e2687284c64576d2B7Cd55393163Fb9f6871ef`
IndexOracleConsumer	`0xd394B8A5dace5EB22933A2a0DCE1C82c1Eb70cd2`
SettlementEngine	`0x68aa3697b1783cb671c575D01C5622c2FA8570e8`

Parameter	Default
Index price scale	0–100
Tick size	0.01
Maximum leverage	5x
Minimum order size	10
Maintenance margin	625 basis points (6.25%)
Liquidation fee	50 basis points (0.5%)
Trading fee	10 basis points (reporting only)
Oracle price staleness limit	60 seconds
Ingestion cycle	30 seconds
Oracle push interval	15 seconds
Stop and bracket check	5 seconds
Internal token lifetime	60 minutes
Faucet	1,000 test USDC per wallet per 24 hours

Disclaimer

This document is for information only. It is not financial, investment, legal or tax advice, and it is not an offer or solicitation to buy or sell any instrument. Xleuth is currently a testnet product that uses test funds with no value. Descriptions of future features are intentions and may change. Prediction markets and leveraged derivatives involve substantial risk, and may be restricted or prohibited in your jurisdiction. Figures and prices in the illustrations are examples only.